

# RISP (Idea 1) — Next Steps, Deferred Items & Costs, and a Frank Quality Review

Companion to EXECUTION\_PLAN.md; status as of the June 11, 2026 build

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## 1. Where the Project Stands

One build session produced the full first-iteration stack: theory with complete proofs (decomposition, eviction-rate dichotomy, KL→regret Pinsker bridge, break-even dormancy), a seeded reproducible codebase (eleven arms, six experiment families, ~1,600 lines, laptop-scale), and three compiled documents—`main.pdf` (19 pp), `RISP Explainer.pdf` (18 pp), `Deep Dive.pdf` (80 pp)—with every number traced to a results JSON and a 46-point numeric audit passed. The headline  $2\times 2$  dissociation is confirmed in controlled markets (retention  $-6.1\%$ ,  $p=1.3\times 10^{-3}$ ; invariance  $-4.4\%$ ,  $p=6.1\times 10^{-3}$ ; super-additive interaction with the predicted inert cell,  $p=0.88$ ; skyline matched at  $p=0.72$ ), and three honest negative results are reported at full prominence (the real-data structure gate fails on available panels; the batching prediction was refuted; the mechanism inverts at high signal-to-noise).

## 2. What To Do Next (priority order)

1. **Submit the WRDS/CRSP request now.** The flagship claim of the roadmap (post-reactivation regret on S&P 500 constituents, 1990–2025, with 2000/2008/2011/2015/2018/2020/2022 as crisis episodes) is the single experiment that decides whether this is an ICAIF/JFDS methodology paper or an AAAI/KDD-grade result. Everything is pre-registered and the code is panel-agnostic; the binding constraint is data-access lead time (typically 2–6 weeks at Penn), not engineering. **[GATE: E0 on equities decides the paper’s final framing (F1 vs. headline-real).]**
2. **Re-run the E0 gate where it might pass, using data already on disk.** Intraday crypto (1H/4H/15m Bybit bars are in `GAUSE/data`) multiplies episode counts and changes the SNR regime; a richer feature inventory (carry, term structure, cross-sectional rank features) is a free upgrade. One week of work, zero dollars. An intraday pass would give a constructive real-data demonstration without WRDS.
3. **Harden the decision layer for generality.** Replace the solver-trivial top- $k$  portfolio with (a) a true knapsack (general budgets/weights) and (b) the execution-scheduling shortest-path task, via Gurobi (free academic license) or OR-Tools (free). This pre-empts the predictable PnO-reviewer objection that linear top- $k$  makes decision-focused learning easy. 1–2 weeks.
4. **The staleness trigger (E6 remedy).** The high-SNR inversion is currently a measured boundary; GAUSE’s staleness trigger (re-open competition on confident underperformance) is the known fix. Implementing it and showing the inversion shrinks would convert the audit’s worst cell into a design recommendation. 1 week.

5. **Scale the statistical battery for submission.** 50–100 seeds on E1–E6 (hours of laptop time), nested walk-forward for  $\beta, \eta, \lambda$ , and—once any real panel passes the gate—the pre-committed DSR/SPA/transaction-cost battery.
6. **Theory polish.** (a) Uniform-convergence version of the episode-transfer lemma for linear heads (covering-number argument; removes the plug-in caveat at the cost of a complexity term); (b) decide the Inv-PnCO coauthor invitation [**GATE: by Jul 1, per roadmap**]  
—it affects framing and theory credibility.
7. **Author read-through of the Deep Dive.** The 80-page document was assembled from parallel-drafted parts; numbers and compilation are audited, but a single careful human pass for seam consistency and voice is the remaining quality step I could not substitute for.
8. **Packaging:** timestamped pre-registration appendix (OSF, free), public repo, ICAIF-26 workshop version as the visibility play while the winter-cycle (AAAI-27/KDD-27) version waits on equities.

### 3. Not Done Due to Expense (with approximate costs)

All prices are *approximate, from memory as of early 2026—verify current pricing before budgeting*. The build deliberately spent \$0.

| Deferred item                                                                         | Approx. cost                                                                         | Notes / cheaper path                                                                                                                                           |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRSP survivorship-bias-free equity panel                                              | <i>\$0 via Penn WRDS;</i><br>commercial <i>\$5k–40k/yr</i>                           | Free path is institutional access; the real cost is 2–6 weeks lead time. This is the highest-value deferred item.                                              |
| Survivorship-aware equity data, no WRDS                                               | <i>\$300–1,200/yr</i><br>(Norgate, Sharadar/Nasdaq Data Link, EODHD tiers)           | Acceptable fallback for the constituents panel; document survivorship handling carefully.                                                                      |
| Cross-asset episodes (credit, FX, futures beyond FRED)                                | <i>\$0–3k/yr</i> free-tier to mid-tier vendors; Bloomberg/Refinitiv <i>\$25k+/yr</i> | The cross-asset invariance upgrade (stronger claim) needs aligned multi-asset panels; G10 FX dailies are nearly free, credit indices are not.                  |
| Neural specialists (small MLP heads) at scale, 100 seeds                              | <i>\$200–500</i> cloud GPU or days on a workstation                                  | Roadmap’s own estimate; linear heads were chosen so the mechanism, not capacity of the function class, drives results—neural replication is a robustness item. |
| Intraday tick/order-book data for execution tasks                                     | <i>\$1k–10k+</i> depending on venue                                                  | Only needed if the execution-scheduling decision layer becomes a headline rather than robustness check.                                                        |
| Large-scale ablation grid (all sweeps $\times$ both memory models $\times$ 100 seeds) | <i>\$0–100</i> (compute time only)                                                   | Deferred for time, not money; do before camera-ready.                                                                                                          |
| Professional copy-edit of the 117 combined pages                                      | <i>\$300–800</i>                                                                     | Optional; the author pass (item 7 above) comes first.                                                                                                          |

Items *not* deferred by cost but by scope discipline: the LLM-agent bridge to Idea 3 (roadmap F5 caps it at one future-work paragraph), and the class-incremental (label-free) variant, which GAUSE showed collapsing under real feature overlap—a research risk, not a line item.

## 4. Frank Quality Review

### 4.1 What is genuinely strong

- **The falsifiable core held.** The  $2 \times 2$  was pre-registered with a distinctive signature—invariance inert in the reward-driven row—that a generic “both tricks help” account would not predict. It confirmed at  $p=0.88$  (inertness) and the per-seed interaction CI excludes zero. Matching the hand-pinned oracle ( $p=0.72$ ) without being given the assignment is the single best result.
- **The honesty infrastructure is real, not decorative.** Two pre-registered predictions were refuted by our own ablations and are reported as findings; the E0 gate failed on real data and the paper ships the null prominently. The E6 inversion (mechanism loses 49% at high SNR) is the kind of result authors usually bury; here it is an audit axis with a measured crossover. This is the part of the work I would defend hardest.
- **Reproducibility discipline.** Every number traces to a seeded JSON; the full battery reruns in under an hour; the soft-memory bug found mid-build was fixed, rerun, and documented rather than papered over.
- **Theory–experiment coupling.** The decomposition’s two knobs map one-to-one onto the experiment’s two axes, and the one place theory and data disagreed (batching) is resolved in the text with the corrected mechanism story.

### 4.2 Weaknesses and risks (the referee’s view)

- **The headline is synthetic, and the real-data result is a null.** This is the central limitation. As it stands the paper is a controlled-study-plus-protocol contribution (roadmap fallback F1), not a demonstrated market result. A top-venue reviewer can accept that framing or reject it as “simulation paper”; the equities gate decides which. *It is too early to know whether the mechanism pays on any real market—that question is empirically open until the intraday and equities E0 runs.*
- **Effect sizes are modest in raw terms.** 10.3% post-reactivation (38% of the closable excess) is honest and statistically solid but far from GAUSE’s +71%; daily-frequency noise floors compress everything. Reviewers anchored on big numbers will need the excess-over-floor framing, which we provide, but it is a framing.
- **The decision layer is solver-trivial.** Linear top- $k$  keeps the PnO structure but the strongest PnO reviewers will ask for a hard combinatorial layer (knapsack, scheduling) before crediting generality. Planned (item 3), not done.
- **Idealized theory.** The plug-in caveat on the transfer lemma, no convergence theorem for the coupled winner-take-all dynamics, and an A3 (capital-allocator) model that practitioners could dispute as stylized. All flagged in-text; none resolved.
- **Single-session authorship of 117 pages.** The Deep Dive’s parts were drafted in parallel and machine-audited for numbers and compilation, but it has not had a cover-to-cover human read; seam-level inconsistencies of voice or emphasis may remain. The main paper at 19 pp also needs restructuring into a 9–10 pp body plus appendix for any NeurIPS/AAAI-format submission.
- **Smaller wrinkles, all disclosed:** the crisis regime never formally pins (affinity 0.67–0.87; ownership is stable, but the pin rule needs a scarce-regime variant); the E1s stitching breaks cross-block dependence; the E0 null is bounded to small universes, daily bars, two labelers, and one feature inventory.

### 4.3 Verdict

The work is a **strong, unusually honest first iteration**—publishable on the methodology-plus-controlled-study framing at ICAIF/JFDS level now, with a credible path to AAAI/KDD/Management-Science level *if and only if* a real panel passes the E0 gate and reproduces even half the synthetic dissociation. The theory, code, protocol, and writing are not the bottleneck; data access is. Whether it becomes the “masterpiece with great academic contribution” of the original brief is therefore not yet knowable, and saying so plainly is part of the standard the project set for itself: the diagnostic-first protocol applies to the project as much as to the markets it studies. Recommended decision sequence: WRDS request today; intraday E0 within two weeks; framing decision (F1 vs. headline-real) when both gates report; coauthor decision Jul 1; venue decision Aug 1.